Hypertext data mining
A tutorial survey

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Hypertext databases

- Academia
  - Digital library, web publication
- Consumer
  - Newsgroups, communities, product reviews
- Industry and organizations
  - Health care, customer service
  - Corporate email

- An inherently collaborative medium
- Bigger than the sum of its parts

The Web

- 2 billion HTML pages, several terabytes
- Highly dynamic
  - 1 million new pages per day
  - Over 600 GB of pages change per month
  - Average page changes in a few weeks
- Largest crawlers
  - Refresh less than 18% in a few weeks
  - Cover less than 50% ever
- Average page has 7–10 links
  - Links form content-based communities

The role of data mining

- Search and measures of similarity
- Unsupervised learning
  - Automatic topic taxonomy generation
- (Semi-) supervised learning
  - Taxonomy maintenance, content filtering
- Collaborative recommendation
  - Static page contents
  - Dynamic page visit behavior
- Hyperlink graph analyses
  - Notions of centrality and prestige

Differences from structured data

- Document ≠ rows and columns
  - Extended complex objects
  - Links and relations to other objects
- Document ≠ XML graph
  - Combine models and analyses for attributes, elements, and CDATA
  - Models different from structured scenario
- Very high dimensionality
  - Tens of thousands as against dozens
  - Sparse: most dimensions absent/irrelevant
- Complex taxonomies and ontologies

The sublime and the ridiculous

- What is the exact circumference of a circle of radius one inch?
- Is the distance between Tokyo and Rome more than 6000 miles?
- What is the distance between Tokyo and Rome?
- java
- java +coffee -applet
- "uninterrupt* power suppl*" ups -parcel

Search products and services

- Verity
- Fulcrum
- PLS
- Oracle text extender
- DB2 text extender
- Infoseek Intranet
- SMART (academic)
- Glimpse (academic)
- Inktomi (HotBot)
- Alta Vista
- Raging Search
- Google
- Dmoz.org
- Yahoo!
- Infoseek Internet
- Lycos
- Excite

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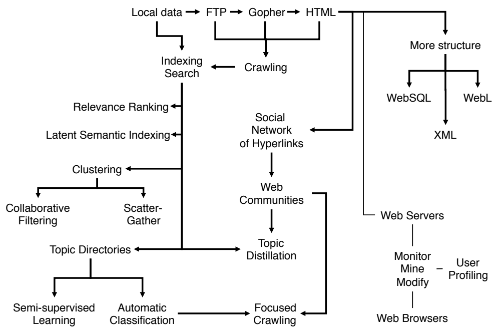

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Roadmap

- Basic indexing and search
- Measures of similarity
- Unsupervised learning or clustering
- Supervised learning or classification
- Semi-supervised learning
- Analyzing hyperlink structure
- Systems issues
- Resources and references

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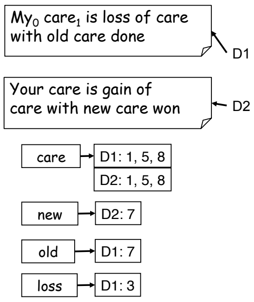
Basic indexing and search

Keyword indexing

- Boolean search
  - care AND NOT old
- Stemming
  - gain*
- Phrases and proximity
  - “new care”
  - loss NEAR/5 care
  - <SENTENCE>



Tables and queries

POSTING

tid	did	pos
care	d1	1
care	d1	5
care	d1	8
care	d2	1
care	d2	5
care	d2	8
new	d2	7
old	d1	7
loss	d1	3
…	…	…

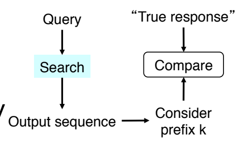
```
select distinct did from POSTING where tid = ‘care’ excep
select distinct did from POSTING where tid like ‘gain%’

with
TPOS1(did, pos) as
        (select did, pos from POSTING where tid = ‘new’ )
TPOS2(did, pos) as
        (select did, pos from POSTING where tid = ‘care’ )
select distinct did from TPOS1, TPOS2
        where TPOS1.did = TPOS2.did
        and proximity(TPOS1.pos, TPOS2.pos)

proximity(a, b) ::=
        a + 1 = b
        abs(a - b) < 5
```

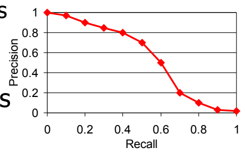
Issues

- Space overhead
  - 5…15% without position information
  - 30…50% to support proximity search
  - Content-based clustering and delta-encoding of document and term ID can reduce space
- Updates
  - Complex for compressed index
  - Global statistics decide ranking
  - Typically batch updates with ping-pong

Relevance ranking

- Recall = coverage
  - What fraction of relevant documents were reported
- Precision = accuracy
  - What fraction of reported documents were relevant
- Trade-off
- 'Query' generalizes to 'topic'





Vector space model and TFIDF

- Some words are more important than others
- W.r.t. a document collection *D*
  - d_+ have a term, d_- do not
  - "Inverse document frequency" $1 + \log \frac{d_+ + d_-}{d+}$
- "Term frequency" (TF)
  - Many variants: $\frac{n(d,t)}{\sum_t n(d,t)}, \frac{n(d,t)}{\max_t n(d,t)}$
- Probabilistic models

‘Iceberg’ queries

- Given a query
  - For all pages in the database computer similarity between query and page
  - Report 10 most similar pages
- Ideally, computation and IO effort should be related to output size
  - Inverted index with AND may violate this
- Similar issues arise in clustering and classification

Similarity and clustering

Clustering

- Given an unlabeled collection of documents, induce a taxonomy based on similarity (such as Yahoo)
- Need document similarity measure
  - Represent documents by TFIDF vectors
  - Distance between document vectors
  - Cosine of angle between document vectors
- Issues
  - Large number of noisy dimensions
  - Notion of noise is application dependent

Document model

- Vocabulary V, term w_i, document α represented by $c(\alpha) = \{f(w_i, \alpha)\}_{w_i \in V}$
- $f(w_i, \alpha)$ is the number of times w_i occurs in document α
- Most f's are zeroes for a single document
- Monotone component-wise damping function g such as log or square-root

$$g(c(\alpha)) = \{g(f(w_i, \alpha))\}_{w_i \in V}$$

Similarity

$$s(\alpha, \beta) = \frac{\langle g(c(\alpha)), g(c(\beta)) \rangle}{\|g(c(\alpha))\| \cdot \|g(c(\beta))\|}$$

$\langle \cdot, \cdot \rangle$ = inner product

Normalized document profile: $p(\alpha) = \frac{g(c(\alpha))}{\|g(c(\alpha))\|}$

Profile for document group Γ: $p(\Gamma) = \frac{\sum_{\alpha \in \Gamma} p(\alpha)}{\left\| \sum_{\alpha \in \Gamma} p(\alpha) \right\|}$

Top-down clustering

- *k*-Means: Repeat…
  - Choose *k* arbitrary 'centroids'
  - Assign each document to nearest centroid
  - Recompute centroids
- Expectation maximization (EM):
  - Pick *k* arbitrary 'distributions'
  - Repeat:
    - Find probability that document *d* is generated from distribution *f* for all *d* and *f*
    - Estimate distribution parameters from weighted contribution of documents

Bottom-up clustering

$$s(\Gamma) = \frac{1}{|\Gamma|(|\Gamma|-1)} \sum_{\alpha\in\Gamma} \sum_{\beta\neq\alpha} s(\alpha,\beta)$$

- Initially G is a collection of singleton groups, each with one document
- Repeat
  - Find Γ, Δ in G with max $s(\Gamma\cup\Delta)$
  - Merge group Γ with group Δ
- For each Γ keep track of best Δ
- $O(n^2 \log n)$ algorithm with n^2 space

Updating group average profiles

Un-normalized group profile: $\hat{p}(\Gamma) = \sum_{\alpha\in\Gamma} p(\alpha)$

Can show:

$$s(\Gamma) = \frac{\langle \hat{p}(\Gamma), \hat{p}(\Gamma)\rangle - |\Gamma|}{|\Gamma|(|\Gamma|-1)}$$

$$s(\Gamma\cup\Lambda) = \frac{\langle \hat{p}(\Gamma\cup\Delta), \hat{p}(\Gamma\cup\Delta)\rangle - (|\Gamma|+|\Delta|)}{(|\Gamma|+|\Delta|)(|\Gamma|+|\Delta|-1)}$$

$$\langle \hat{p}(\Gamma\cup\Delta), \hat{p}(\Gamma\cup\Delta)\rangle = \langle \hat{p}(\Gamma), \hat{p}(\Gamma)\rangle + \langle \hat{p}(\Delta), \hat{p}(\Delta)\rangle + 2\langle \hat{p}(\Gamma), \hat{p}(\Delta)\rangle$$

"Rectangular time" algorithm

- Quadratic time is too slow
- Randomly sample $O(\sqrt{kn})$ documents
- Run group average clustering algorithm to reduce to k groups or clusters
- Iterate assign-to-nearest $O(1)$ times
  - Move each document to nearest cluster
  - Recompute cluster centroids
- Total time taken is $O(kn)$
- Non-deterministic behavior

Issues

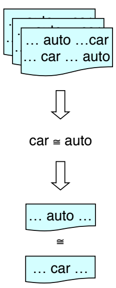
- Detecting noise dimensions
  - Bottom-up dimension composition too slow
  - Definition of noise depends on application
- Running time
  - Distance computation dominates
  - Random projections
  - Sublinear time w/o losing small clusters
- Integrating semi-structured information
  - Hyperlinks, tags embed similarity clues
  - A link is worth a ⌧⌧⌧?⌧⌧⌧ words

Random projection

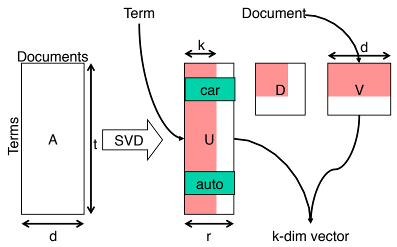
- Johnson-Lindenstrauss lemma:
  - Given a set of points in n dimensions
  - Pick a randomly oriented k dimensional subspace, k in a suitable range
  - Project points on to subspace
  - Inter-point distance is preserved w.h.p.
- Preserve sparseness in practice by
  - Sampling original points uniformly
  - Pre-clustering and choosing cluster centers
  - Projecting other points to center vectors

Extended similarity

- Where can I fix my scooter?
- A great garage to repair your 2-wheeler is at …
- auto and car co-occur often
- Documents having related words are related
- Useful for search and clustering
- Two basic approaches
  - Hand-made thesaurus (WordNet)
  - Co-occurrence and associations



Latent semantic indexing



LSI summary

- SVD factorization applied to term-by-document matrix
- Singular values with largest magnitude retained
- Linear transformation induced on terms and documents
- Documents preprocessed and stored as LSI vectors
- Query transformed at run-time and best documents fetched

Collaborative recommendation

- People=record, movies=features
- People and features to be clustered
  - Mutual reinforcement of similarity
- Need advanced models

	Batman	Rambo	Andre	Hiver	Whispers	StarWars
Lyle						
Ellen						
Jason						
Fred						
Dean						
Karen						

From *Clustering methods in collaborative filtering*, by Ungar and Foster

A model for collaboration

- People and movies belong to unknown classes
- P_k = probability a random person is in class k
- P_l = probability a random movie is in class l
- P_{kl} = probability of a class-k person liking a class-l movie
- Gibbs sampling: iterate
  - Pick a person or movie at random and assign to a class with probability proportional to P_k or P_l
  - Estimate new parameters

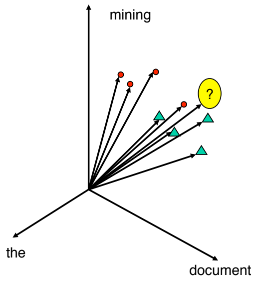
Supervised learning

Supervised learning (classification)

- Many forms
  - Content: automatically organize the web per Yahoo!
  - Type: faculty, student, staff
  - Intent: education, discussion, comparison, advertisement
- Applications
  - Relevance feedback for re-scoring query responses
  - Filtering news, email, etc.
  - Narrowing searches and selective data acquisition

Nearest neighbor classifier

- Build an inverted index of training documents
- Find *k* documents having the largest TFIDF similarity with test document
- Use (weighted) majority votes from training document classes to classify test document



Difficulties

- Context-dependent noise (taxonomy)
  - 'Can' (v.) considered a 'stopword'
  - 'Can' (n.) may not be a stopword in /Yahoo/SocietyCulture/Environment/ Recycling
- Dimensionality
  - Decision tree classifiers: dozens of columns
  - Vector space model: 50,000 'columns'
  - Computational limits force independence assumptions; leads to poor accuracy

Techniques

- Nearest neighbor
  + Standard keyword index also supports classification
  – How to define similarity? (TFIDF may not work)
  – Wastes space by storing individual document info
- Rule-based, decision-tree based
  – Very slow to train (but quick to test)
  + Good accuracy (but brittle rules tend to overfit)
- Model-based
  + Fast training and testing with small footprint
- Separator-based
  * Support Vector Machines

Document generation models

- Boolean vector (word counts ignored)
  - Toss one coin for each term in the universe
- Bag of words (multinomial)
  - Toss coin with a term on each face
- Limited dependence models
  - Bayesian network where each feature has at most *k* features as parents
  - Maximum entropy estimation
- Limited memory models
  - Markov models

Binary (boolean vector)

- Let vocabulary size be $|T|$
- Each document is a vector of length $|T|$
  - One slot for each term
- Each slot t has an associated coin with head probability ϕ_t
- Slots are turned on and off independently by tossing the coins

$$\Pr(d \mid c) = \prod_{t \in d} \phi_{c,t} \prod_{t \notin d} (1 - \phi_{c,t})$$

Multinomial (bag-of-words)

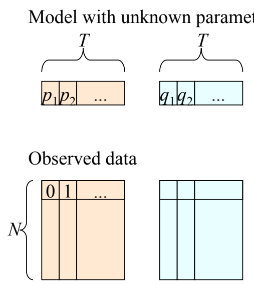
- Decide topic; topic c is picked with prior probability $\pi(c)$; $\sum_c \pi(c) = 1$
- Each topic c has parameters $\theta(c,t)$ for terms t
- Coin with face probabilities $\sum_t \theta(c,t) = 1$
- Fix document length ℓ
- Toss coin ℓ times, once for each word
- Given ℓ and c, probability of document

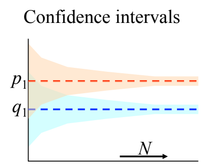
$$\Pr[d \mid c, n(d) = \ell] = \binom{n(d)}{\{n(d,t)\}} \prod_{t \in d} \theta(c,t)^{n(d,t)}$$

Limitations

- With the term distribution
  - 100th occurrence is as surprising as first
  - No inter-term dependence
- With using the model
  - Most observed $\theta(c,t)$ are zero and/or noisy
  - Have to pick a low-noise subset of the term universe
  - Have to “fix” low-support statistics
    - Smoothing and discretization
    - Coin turned up heads 100/100 times; what is Pr(tail) on the next toss?

Feature selection

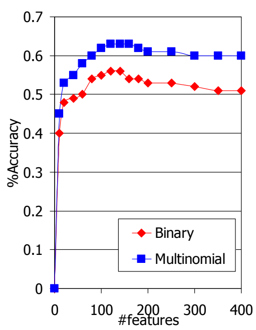




Pick $F \subseteq T$ such that models built over F have high separation confidence

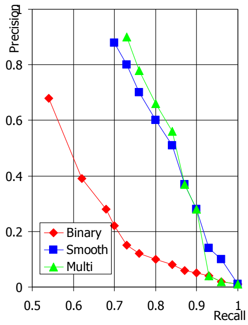
Effect of feature selection

- Sharp knee in error with small number of features
- Saves class model space
  - Easier to hold in memory
  - Faster classification
- Mild increase in error beyond knee
  - Worse for binary model



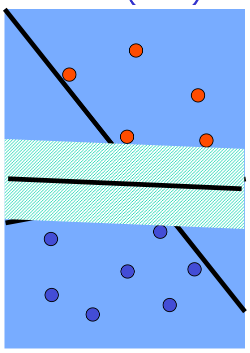
Effect of parameter smoothing

- Multinomial known to be more accurate than binary under Laplace smoothing
- Better marginal distribution model compensates for modeling term counts!
- Good parameter smoothing is critical



Support vector machines (SVM)

- No assumptions on data distribution
- Goal is to find separators
- Large bands around separators give better generalization
- Quadratic programming
- Efficient heuristics
- Best known results



Maximum entropy classifiers

- Observations (d_i, c_i), $i = 1 \ldots N$
- Want model $p(c\,|\,d)$, expressed using features $f_i(c, d)$ and parameters λ_j as

$$p(c\,|\,d) = \frac{1}{Z(d)} \prod_j e^{\lambda_j f_j(c,d)}, Z(d) = \sum_{c'} p(c'\,|\,d)$$

- Constraints given by observed data

$$\sum_{d,c} \tilde{p}(d) p(c\,|\,d) f(d,c) = \sum_{d,c} \tilde{p}(d,c) f(d,c)$$

- Objective is to maximize entropy of p

$$H(p) = -\sum_{d,c} \tilde{p}(d) p(c\,|\,d) \log p(c\,|\,d)$$

- Features
  - Numerical non-linear optimization
  - No naïve independence assumptions

Semi-supervised learning

Exploiting unlabeled documents

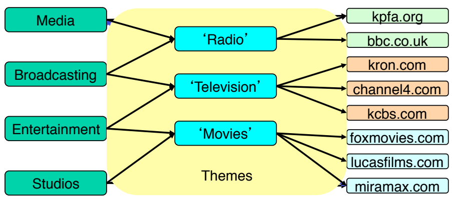
- Unlabeled documents are plentiful; labeling is laborious
- Let training documents belong to classes in a *graded* manner Pr(*c*|*d*)
- Initially labeled documents have 0/1 membership
- Repeat (Expectation Maximization 'EM'):
  - Update class model parameters θ
  - Update membership probabilities Pr(*c*|*d*)
- Small labeled set→large accuracy boost

Clustering categorical data

- Example: Web pages bookmarked by many users into multiple folders
- Two relations
  - Occurs_in(term, document)
  - Belongs_to(document, folder)
- Goal: cluster the documents so that original folders can be expressed as simple union of clusters
- Application: user profiling, collaborative recommendation

Bookmarks clustering

- Unclear how to embed in a geometry
  - A folder is worth __?__ words?
- Similarity clues: document-folder cocitation and term sharing across folders



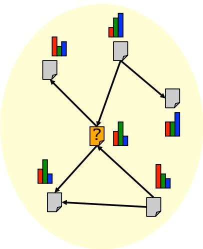
Analyzing hyperlink structure

Hyperlink graph analysis

- Hypermedia is a **social network**
  - Telephoned, advised, co-authored, paid
- Social network theory (cf. Wasserman & Faust)
  - Extensive research applying graph notions
  - **Centrality and prestige**
  - **Co-citation (relevance judgment)**
- Applications
  - Web search: HITS, Google, CLEVER
  - Classification and topic distillation

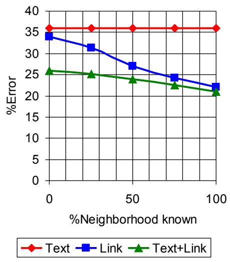
Hypertext models for classification

- *c*=class, *t*=text, *N*=neighbors
- Text-only model: Pr[*t*| *c*]
- Using neighbors' text to judge my topic: Pr[*t*, *t*(*N*) | *c*]
- Better model: Pr[*t*, *c*(*N*) | *c*]
- Non-linear relaxation



Exploiting link features

- 9600 patents from 12 classes marked by USPTO
- Patents have text and cite other patents
- Expand test patent to include neighborhood
- 'Forget' fraction of neighbors' classes



Co-training

- Divide features into two class-conditionally independent sets
- Use labeled data to induce two separate classifiers
- Repeat:
  - Each classifier is "most confident" about some unlabeled instances
  - These are labeled and added to the training set of the other classifier
- Improvements for text + hyperlinks

Ranking by popularity

- In-degree ≈ prestige
- Not all votes are worth the same
- Prestige of a page is the sum of prestige of citing pages:
  $\boldsymbol{p} = \boldsymbol{Ep}$
- Pre-compute query independent prestige score
- Google model
- High prestige ⇔ good authority
- High reflected prestige ⇔ good hub
- Bipartite iteration
  - $\boldsymbol{a} = \boldsymbol{Eh}$
  - $\boldsymbol{h} = \boldsymbol{E}^T\boldsymbol{a}$
  - $\boldsymbol{h} = \boldsymbol{E}^T\boldsymbol{Eh}$
- HITS/Clever model

Tables and queries

HUBS

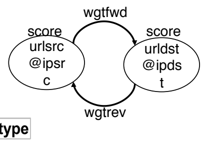
url	score

AUTH

url	score

```
delete from HUBS;
insert into HUBS(url, score)
        (select urlsrc, sum(score * wtrev) from AUTH, LINK
        where authwt is not null and type = non-local
        and ipdst <> ipsrc and url = urldst
        group by urlsrc);
update HUBS set (score) = score /
        (select sum(score) from HUBS);
```

```
update LINK as X set (wtfwd) = 1. /
        (select count(ipsrc) from LINK
        where ipsrc = X.ipsrc
        and urldst = X.urldst)
        where type = non-local;
```

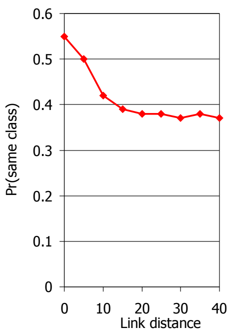


LINK

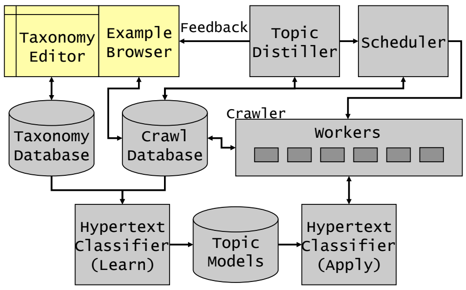
urlsrc	urldst	ipsrc	ipdst	wgtfwd	wtrev	type

Topical locality on the Web

- Sample sequence of out-links from pages
- Classify out-links
- See if class is same as that at offset zero
- TFIDF similarity across endpoint of a link is very large compared to random page-pairs

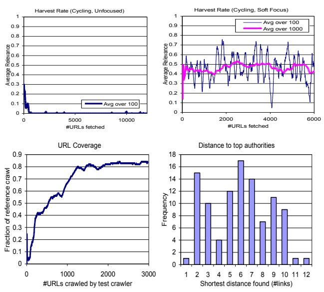


Resource discovery



Resource discovery results

- High rate of "harvesting" relevant pages
- Robust to perturbations of starting URLs
- Great resources found 12 links from start set



Systems issues

Data capture

- Early hypermedia visions
  - Xanadu (Nelson), Memex (Bush)
  - Text, links, browsing and searching actions
- Web as hypermedia
  - Text and link support is reasonable
    - Autonomy leads to some anarchy
  - Architecture for capturing user behavior
    - No single standard
    - Applications too nascent and diverse
    - Privacy concerns

Storage, indexing, query processing

- Storage of XML objects in RDBMS is being intensively researched
- Documents have unstructured fields too
- Space- and update-efficient string index
  - Indices in Oracle8i exceed 10x raw text
- Approximate queries over text
- Combining string queries with structure queries
- Handling hierarchies efficiently

Concurrency and recovery

- Strong RDBMS features
  - Useful in medium-sized crawlers
- Not sufficiently flexible
  - Unlogged tables, columns
  - Lazy indices and concurrent work queues
- Advances query processing
  - Index (-ed scans) over temporary table expressions; multi-query optimization
  - Answering complex queries approximately

Resources

Research areas

- Modeling, representation, and manipulation
- Approximate structure and content matching
- Answering questions in specific domains
- Language representation
- Interactive refinement of ill-defined queries
- Tracking emergent topics in a newsgroup
- Content-based collaborative recommendation
- Semantic prefetching and caching

Events and activities

- Text REtrieval Conference (TREC)
  - Mature ad-hoc query and filtering tracks
  - New track for web search (2…100GB corpus)
  - New track for question answering
- Internet Archive
  - Accounts with access to large Web crawls
- DIMACS special years on Networks (-2000)
  - Includes applications such as information retrieval, databases and the Web, multimedia transmission and coding, distributed and collaborative computing
- Conferences: WWW, SIGIR, KDD, ICML, AAAI